RAK18001 WisBlock Buzzer Module Datasheet

Overview

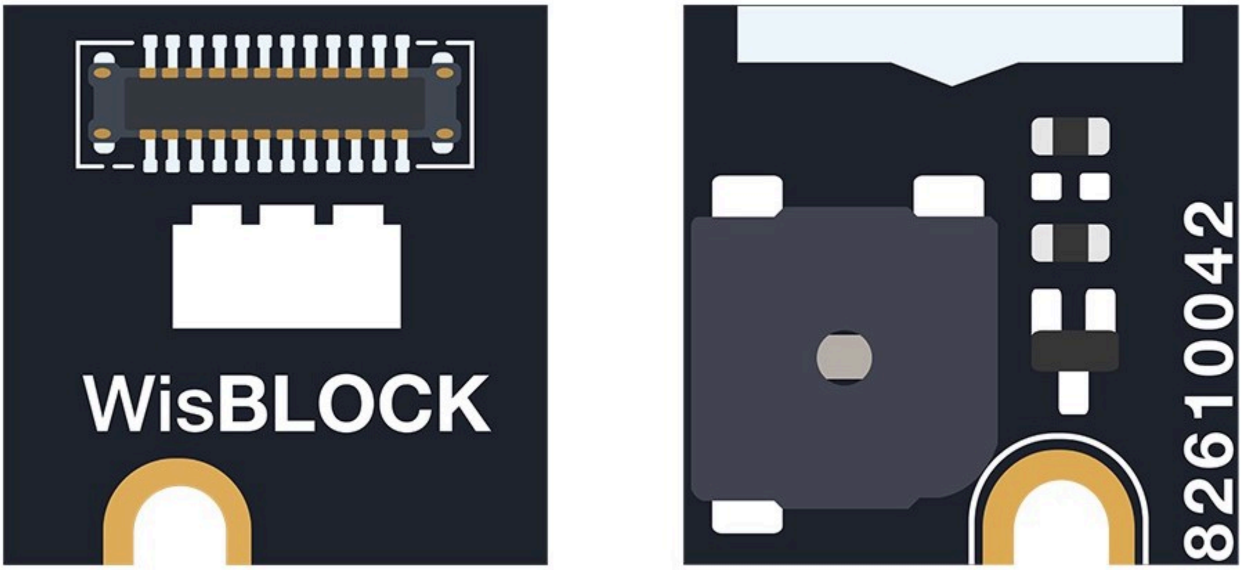


Figure 1: RAK18001 Buzzer Module

Description

The **RAK18001** is a WisBlock Extra module that uses a **MLT-5020** as its built-in buzzer. It produces an audible high-pitched sound which can be used in various alarm and notifier applications. The sound and loudness can be controlled through PWM (Pulse-Width Modulation) signal from a WisBlock Core. The output sounds and pitch level is customizable to the point that it is even possible to play a melody.

Features

- 3.3 V Input Voltage: On/Off Control by the WisBlock Core
- Operating Voltage: 2.0 V - 5.0 V
- 75 dB sound output at 10 cm distance
- Resonant Frequency: 4000 Hz

- PWM Controlled: Loudness and pitch level can be customized using your code
- Uses MLT-5020 Buzzer - small and compact
- Chipset: Jiangsu Huaneng Electronics MLT-5020
- Module Size: 10 mm x 10 mm
- Built-in Buzzer Size: 5 mm x 5 mm x 2 mm

Specifications

Overview

Mounting

The RAK18001 Buzzer Module can be mounted to any IO Slots (A, B, C, or D) of WisBlock Base Board. **Figure 2** shows the mounting mechanism of the RAK18001 on a WisBlock Base, such as the RAK5005-O.

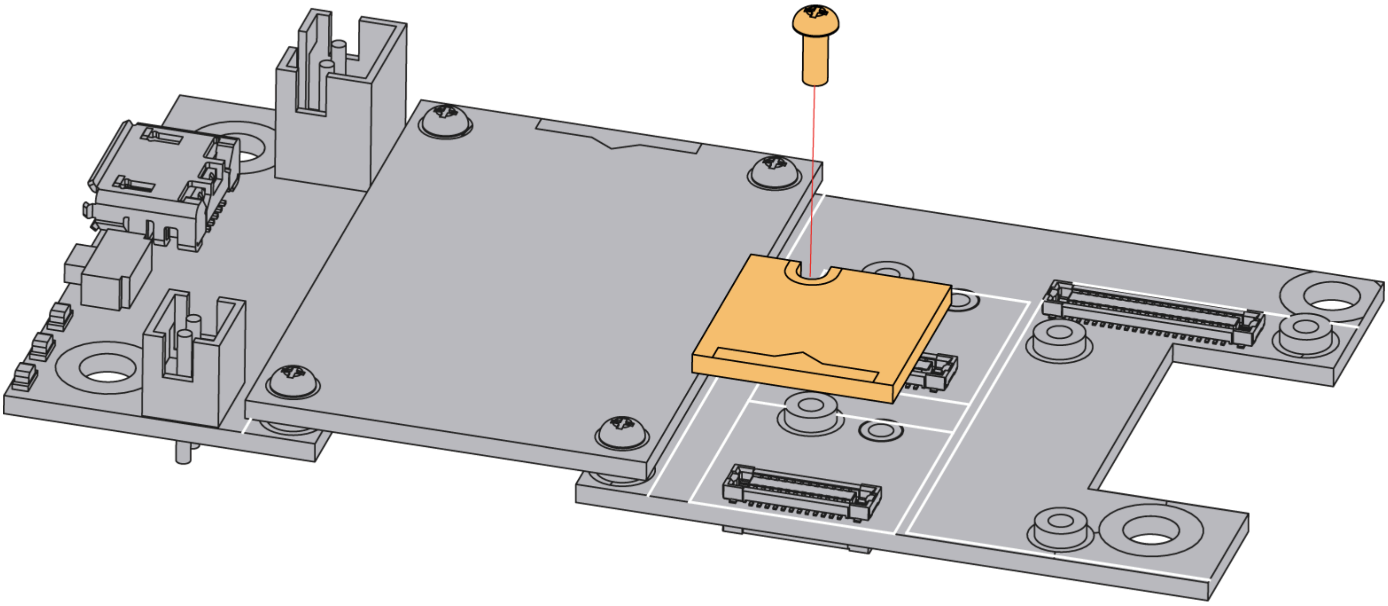


Figure 2: RAK18001 WisBlock Buzzer Module Mounting

Hardware

The hardware specification is categorized into five parts. It shows the chipset of the module and discusses the pinouts, and the corresponding functions and diagrams. It also covers the electrical and mechanical parameters, including the tabular data of the functionalities and standard values of the RAK18001 WisBlock Buzzer Module Datasheet.

Chipset

The RAK18001 uses the MLT-5020 Buzzer.Refer to the manufacturer's MLT-5020 Buzzer Page for more details.

Vendor	Part Number
Jiangsu Huaneng Electronics	MLT-5020

Pin Definition

The RAK18001 WisBlock Buzzer Module comprises a standard WisSensor connector. The WisSensor connector allows the RAK18001 module to be mounted on a WisBlock baseboard, such as RAK5005-O. The pin order of the connector and the pinout definition are shown in **Figure 3**.

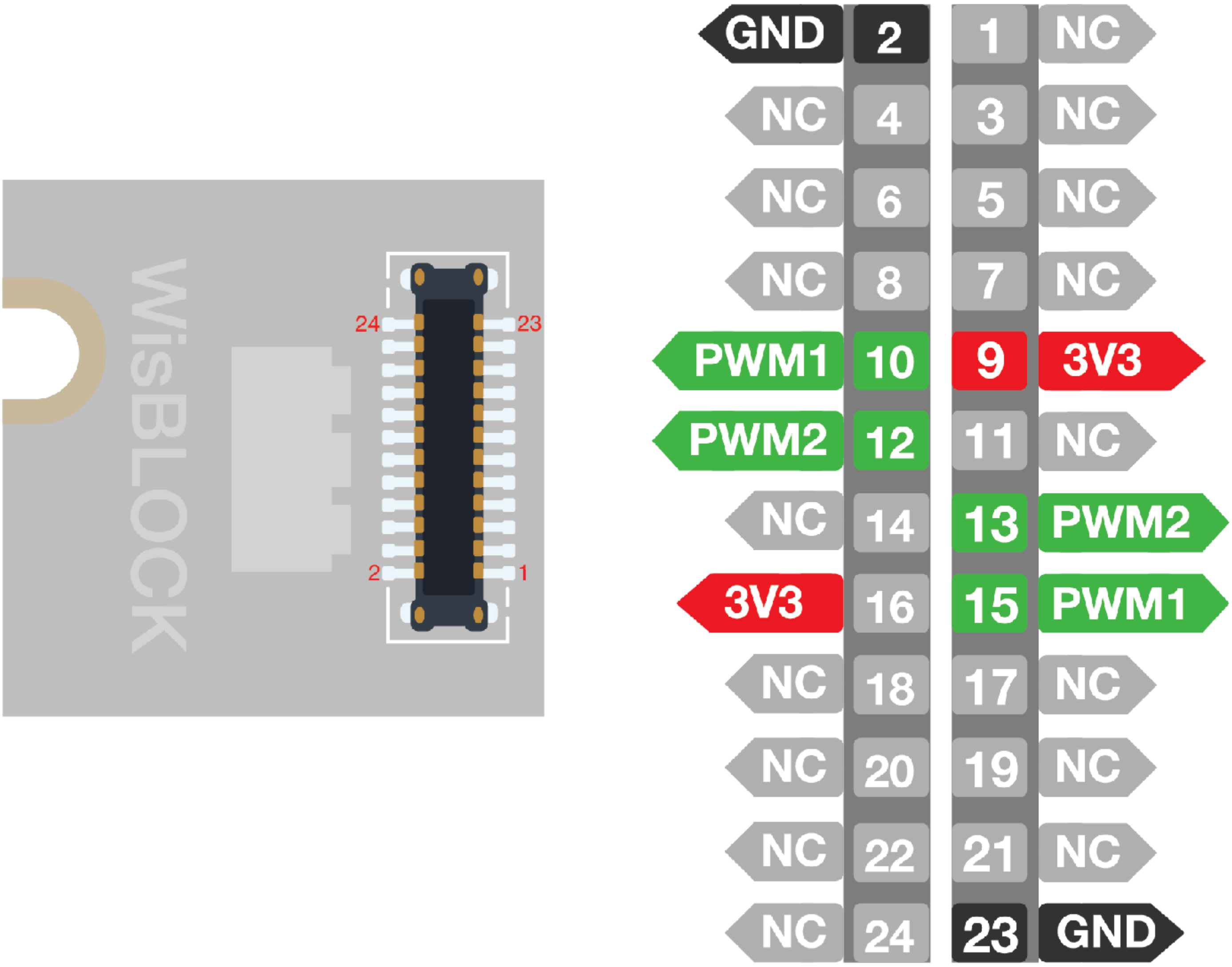


Figure 3: RAK18001 Buzzer Module Pinout Diagram

The table below shows the default IOs used for different slots:

SLOT A	SLOT B	SLOT C	SLOT D
IO1	IO2	IO3	IO5

As seen on **Figure 2**, the PWM1 is connected to R5 which is an NC (no connection). On the other hand, PWM2 is connected to a 330 Ω resistor R6.

However, you can switch these two terminals by connecting the PWM1 to R6 and PWM2 to NC. This configuration will create an alternative IO pins as shown below:

SLOT A	SLOT B	SLOT C	SLOT D
IO2	IO1	IO4	IO6

WARNING

Make sure to set the PWM pin to ***LOW*** after playing a sound. This is to ensure that the buzzer on the RAK18001 is in complete shut down and does not get hot.

Electrical Characteristics

The table below shows the RAK18001 Buzzer Module electrical characteristics:

Symbol	Description	Min.	Nom.	Max.	Unit
V_{DD}	Power supply for the module	2.7	3.3	3.6	V
I_{stb}	Standby current	-	-	5	µA
I_{DD}	Measure current (normal mode)				µA
I_{sd}	Buzzer turn off current				µA

Mechanical Characteristics

Board Dimensions

Figure 4 shows the dimensions and mechanical drawing of the RAK18001 module.

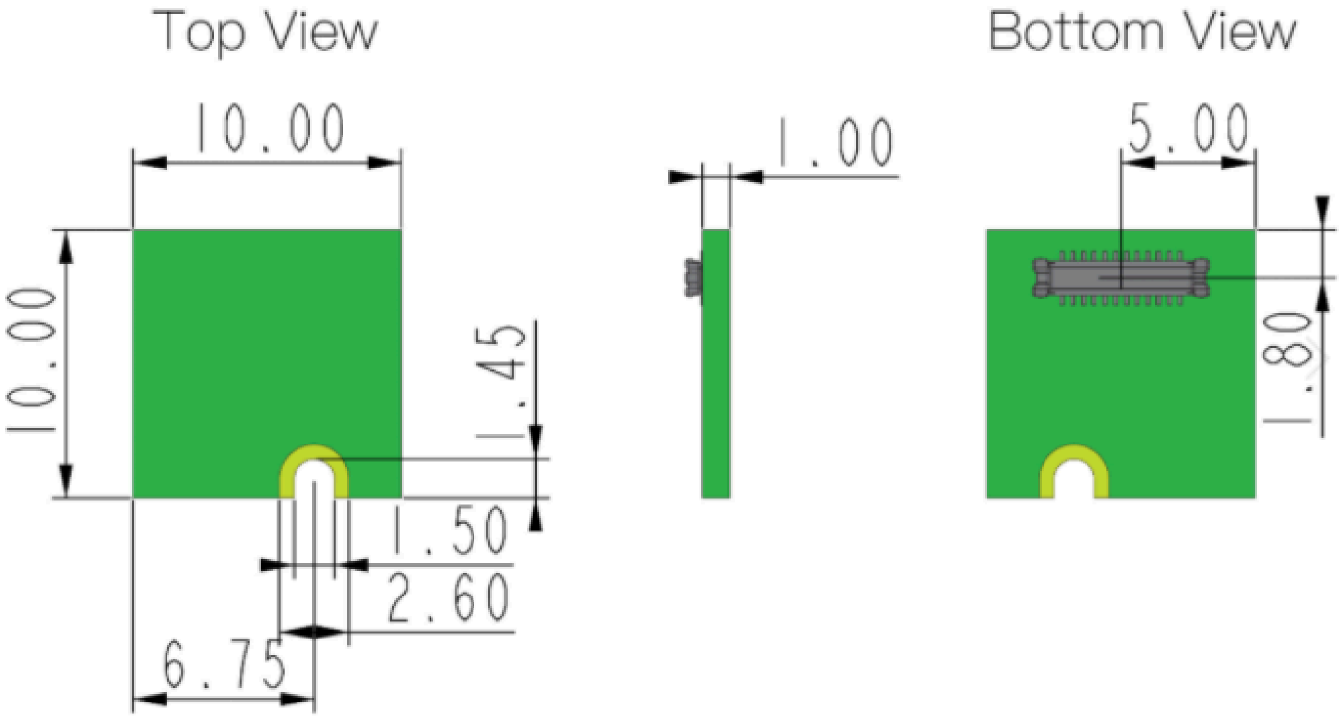
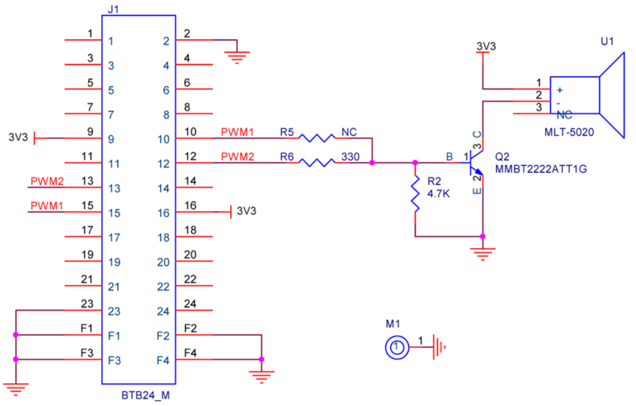


Figure 4: RAK18001 Buzzer Module Mechanical Characteristics

Schematic Diagram

Figure 5 shows the schematic diagram of RAK18001 Module. It consist of the pinouts of both RAK18001 Module and the Built-in Buzzer (MLT-5020).



Electro-Magnetic Buzzer

Rated Voltage: 3.0V

Operating Voltage: 2.0 - 5.0V

Sound Output at 10cm: 75dB

Resonant Frequency: 4000Hz

Dimensions LxWxH (mm): 5 x 5 x 2

Figure 5: RAK18001 Buzzer Module Schematic